Fig. 8: Fibre-optic sensor capsule

fully-open or fully-closed. Proportional solenoid valves, however, operate with a proportional action. By varying the electrical input to a proportional valve, the flow of water through the valve can be continuously and steplessly adjusted between zero and 100 per cent of the maximum rated flow. In order to obtain the proportional action, the solenoid operator is modified.

Proportional solenoid valves operate on the principle that the pull force produced by an electromagnetic coil increases as coil current (Is) is increased, while, likewise, the counter-acting spring force increases when the top spring is compressed.

(a) If the current is less than Io*, preset tension of spring Fs is greater than electromagnetic pull force Fm and the valve remains closed.

(b) If current increases to above Io, Fm is greater than Fs and the core starts to move. Movement of the core compresses the spring, causing Fs to increase. This continues until the new values of Fs and Fm are once again in balance. The process continues so that at any value of I, the core moves into a position where Fs and Fm are balanced. Specific parts of a proportional solenoid valve are shown in Fig. 7.

[*A proportional solenoid valve can be made to open or close infinitely by regulating the current (Is) flowing through the solenoid coil. Coil current (Is) is mainly regulated by regulating the voltage (Us) across the coil. Voltage across the coil can be derived from various supplies, for example, a straight DC supply or a pulse-width modulated supply.

In order to maintain the valve in a specific position, coil current (Is) should be kept substantially independent of changes in the coil-winding resistance due to temperature variations (caused by power input and ambient temperature) typically between 100mA (Io) and 500mA (Imax).]

6. Both the sensor/controller electronics as well as the solenoid valve require a power source. Often, standard battery packs are used as the power source. Commonly-used batteries are C, AA, 6V and 9V lithium batteries. Even so, automatic faucets using an AC transformer as the power source are relatively cheaper than their battery-powered counterparts.

7. Most IR faucet spouts (for water delivery) are designed to house the IR sensor module within these, or fibre-optic sensor capsule to carry the IR signal from the sensor to the spout. Some trendy spouts house the whole sensor, control electronics, solenoid valve and the AC/DC power supply unit within these.

IR water faucets reflect a great combination of innovative design and advanced technology. There are many models in modern designs that go well with other modern accessories and fittings for bathrooms. Like all traditional faucets, you can get IR water faucets in multiple designs and materials.

Generally, IR water faucets are easy to care for and need little maintenance. If you use a type with batteries, you will need to change these when these run out of power. Electronic components can fail in rare cases, but you can simply replace the whole electronic assembly at once, if required. Apart from some little issues that might come up every few years, if these do at all, IR water faucets should not require any further maintenance. EFY